

Lymphoid Neoplasms - Factoid Sheet

Week 7 Hematology • Lecture 7 • Savage • memorize-first companion

ScholarRx: Hematopoietic Malignancies • Chronic Lymphoproliferative Disorders • Lymphoma • Putting It All Together

For every lymphoma: cell/stage -> site -> morphology -> markers -> genetic lesion -> indolent vs aggressive.

Precursor / Hodgkin essentials

Disease	Cold facts
B-ALL/LBL	Child; marrow/blood; TdT+, CD34+, CD19+, CD10+; t(12;21) favorable in deck, t(9;22) unfavorable.
T-ALL/LBL	Teen male + mediastinal/thymic mass; TdT+, CD3/CD7+.
Classic Hodgkin	RS/owl-eye; CD15+ , CD30+ , CD45- ; contiguous nodal spread; often above diaphragm.
NLPHL	Popcorn/L&H cells; CD20+ , CD45+ , CD15-/CD30- ; indolent.

Diagnostic tools

- **IHC**: markers in tissue + architecture
- **Flow**: co-expression/clonality in cell suspension
- **FISH**: rearrangements/copy number
- **Light-chain restriction**: monoclonality clue

HL vs NHL

- **HL**: contiguous spread, RS cells, extranodal less common
- **NHL**: often multicentric/disseminated, extranodal common

Mature B-cell fingerprint table

Disease	Marker / genetics	Classic clue
CLL/SLL	CD5+, CD23+ , dim CD20	Old; smudge cells; AIHA; hypogammaglobulinemia
Hairy cell	BRAF V600E; CD11c/CD25/CD103; TRAP+	Splenomegaly + pancytopenia + dry tap
Follicular	t(14;18) BCL2 ; CD10+	Indolent; uniform follicles; no tingible-body macrophages
MALT	Chronic antigen stimulation	H. pylori stomach; Sjögren; Hashimoto; treat H. pylori
Mantle	t(11;14) cyclin D1 ; CD5+, CD23-	Older male; GI; more aggressive
DLBCL	Large B cells; variable MYC/BCL2/BCL6	Rapidly enlarging mass; aggressive but potentially curable
Burkitt	t(8;14) MYC ; CD10+; Ki-67 ~100%	Starry sky; jaw/endemic or abdominal/sporadic

The translocation trio: 14;18 = follicular/BCL2 • 11;14 = mantle/cyclin D1 • 8;14 = Burkitt/MYC.

T-cell + Plasma Cell + Vignette Factoids

MF / Sézary • Cutaneous CD4 T-cell lymphoma • Pautrier microabscess • Cerebriform nuclei • CD3+, CD4+, CD7 loss • Sézary = erythroderma + blood involvement	Plasma-cell phenotype • CD138+, clonal immunoglobulin • M-spike / paraprotein • Rouleaux • Lytic lesions • Normal Ig suppressed -> infections
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MGUS vs smoldering vs active myeloma

State	Core logic
MGUS	Low clonal burden; M-protein may be present; no myeloma-defining event; classic companion threshold <10% marrow plasma cells.
Smoldering	Higher clonal burden than MGUS but no myeloma-defining event.
Active myeloma	Clonal plasma cells/plasmacytoma + CRAB or SLiM myeloma-defining event.

CRAB: hyperCalcemia • Renal injury • Anemia • Bone lesions.
SLiM enrichment: ≥60% marrow plasma cells • FLC ratio ≥100 (involved FLC ≥100 mg/L) • ≥2 MRI focal lesions.

AL amyloid

Monoclonal **light-chain** protein deposits in tissues. Think nephrotic syndrome, restrictive cardiomyopathy, macroglossia, periorbital purpura. Congo red -> apple-green birefringence.

If you see X -> think Y

• TdT+ child + pancytopenia -> B-ALL • TdT+ teen + mediastinal mass -> T-ALL/LBL • CD15/30 + lacunar cells/fibrous bands -> nodular sclerosis HL • CD5+ CD23+ + smudge -> CLL • CD5+ CD23- cyclin D1+ -> mantle	• t(14;18) + painless nodes -> follicular • H. pylori + gastric lymphoid mass -> MALT • Rapid mass + large B cells -> DLBCL • t(8;14) + starry sky -> Burkitt • Back pain + M-spike + lytic lesions -> myeloma
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Last 5 minutes: TdT=blast. Classic HL=CD15/30. CLL=5+23+. Mantle=5+23-, cyclin D1, 11;14. Follicular=14;18 BCL2. Burkitt=8;14 MYC. MF=Pautrier. Myeloma=CRAB/SLiM + monoclonal plasma cells.

Primary source: Savage 70-slide Lymphoid Neoplasms deck. Official Week 7 LOs control emphasis. HCL, NLPHL, ancillary-test framework, and selected board details are explicit companion gap-fills.